

From Graphical Models to Generative AI

Exact Dynamic Programming and Sum-Product on Trees

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Lecture 4 — Wednesday, September 2, 2026
Mathematics 514

Chapter 3: elimination, separators, messages, and why trees are exceptional.

Research studios and the Project Card

Continuous Structured GenAI

scientific trajectories and fields; imaging; molecular/protein ensembles; diffusion, sampling, steering, constraints

Quantum / Stat Phys / Information

quantum states; spins; random media; BP, loops, tensors, statistical-physics structure

Optimization / Networked Systems

graph/network optimization; power systems; min-max/geometric optimization; constraints and guarantees

Separate studio guide on D2L: book map + suggested papers. Project Card due Sep. 11.

Individual projects; studios are research neighborhoods. Students are welcome to bring relevant papers beyond the suggested lists.

From Lecture 3: dynamic programming on a chain

For a four-variable chain,

$$Z = \sum_{x_1, x_2, x_3, x_4} \psi_1(x_1) \psi_{12}(x_1, x_2) \psi_2(x_2) \psi_{23}(x_2, x_3) \psi_3(x_3) \psi_{34}(x_3, x_4) \psi_4(x_4).$$

Pull the sums inward using the distributive law:

$$m_{1 \rightarrow 2}(x_2) = \sum_{x_1} \psi_1(x_1) \psi_{12}(x_1, x_2),$$

$$m_{2 \rightarrow 3}(x_3) = \sum_{x_2} \psi_2(x_2) \psi_{23}(x_2, x_3) m_{1 \rightarrow 2}(x_2),$$

$$m_{3 \rightarrow 4}(x_4) = \sum_{x_3} \psi_3(x_3) \psi_{34}(x_3, x_4) m_{2 \rightarrow 3}(x_3),$$

$$Z = \sum_{x_4} \psi_4(x_4) m_{3 \rightarrow 4}(x_4).$$

A message summarizes the eliminated graph on the separator still visible to the surviving graph.

One exact operation: eliminate a variable

$$g_i(\mathbf{x}_{S_i}) = \sum_{x_i} \prod_{a: i \in \partial a} f_a(\mathbf{x}_a), \quad S_i = \left(\bigcup_{a: i \in \partial a} \partial a \right) \setminus \{i\}.$$

collect $\rightarrow$ combine $\rightarrow$ eliminate $\rightarrow$ replace

Exactness

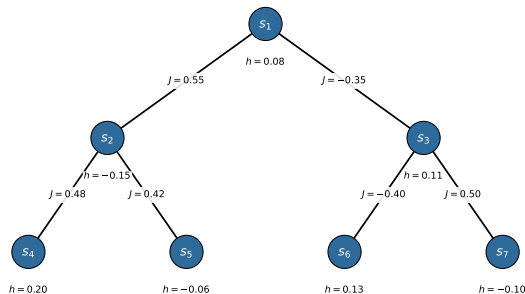
The algebraic identity is exact for every elimination order.

Complexity

The size of the separator S_i controls the intermediate table.

Why trees are exceptional

Binary tree model; node and edge labels are local parameters



Cut one edge (i, j) :

tree $\implies$ two disjoint components.

A message across the cut needs to remember only the boundary variable.

On a tree, messages are exact partial partition functions.

The pairwise tree message

For

$$p(\mathbf{x}) \propto \prod_i \psi_i(x_i) \prod_{(i,j) \in E} \psi_{ij}(x_i, x_j),$$

cut edge (i, j) and summarize the component on the i side:

$$m_{i \rightarrow j}(x_j) = \sum_{x_i} \psi_i(x_i) \psi_{ij}(x_i, x_j) \prod_{k \in \partial i \setminus \{j\}} m_{k \rightarrow i}(x_i)$$

One inward pass + one outward pass = all directed messages.

Beliefs and the partition function

Singleton marginal

$$b_i(x_i) \propto \psi_i(x_i) \prod_{k \in \partial i} m_{k \rightarrow i}(x_i).$$

On a tree: $b_i = p_i$ exactly.

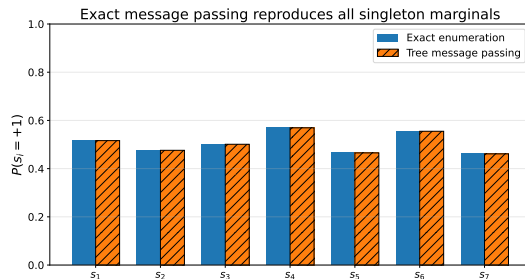
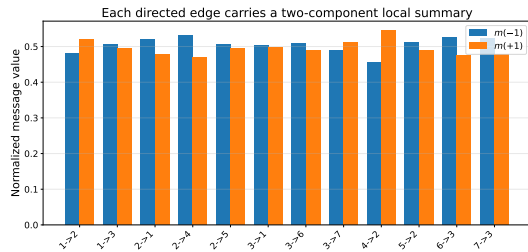
Partition function at a root r

$$Z = \sum_{x_r} \psi_r(x_r) \prod_{k \in \partial r} \tilde{m}_{k \rightarrow r}(x_r),$$

with unnormalized
partial-partition-function messages.

The root changes the schedule, not the probability law or the final answer.

Notebook laboratory: exact messages versus exact enumeration



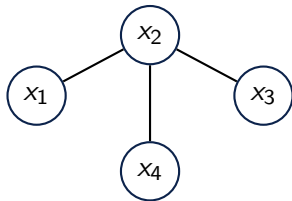
ch03/ch03_02_sum_product_on_trees.ipynb

Same topology, different algebra: dynamic programming is broader than probability

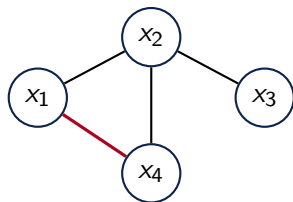
Task	combine	eliminate
Partition function / marginals	$\times$	$\sum$
MAP value	$+$ in log domain	max
Shortest path	$+$	min
Boolean feasibility	$\wedge$	$\vee$

Tree topology controls exact locality; the algebra controls which computational task is solved.

Add one edge: what exactly is lost?



tree: disjoint subproblems



loop: information can return by another route

The local BP equations can still be iterated — but the partial-partition-function proof is gone.